Answer key: Lesson 3, Doubles and Near Doubles

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Look at the dominoes. Fill in the missing numbers.

Answer: 6, 8, 10

Each domino has two halves with the same number of dots. 3 + 3 = 6. 4 + 4 = 8. 5 + 5 = 10.

Watch for: Counting only one half of the domino.; Writing 5 + 5 = 55 (joining the digits).

S2. Mei has 6 red counters and 6 yellow counters. Fill in the missing numbers.

Answer: 6 + 6 = 12; double 6 is 12

The first frame has 6 red counters and the second has 6 yellow counters. 6 + 6 = 12, so double 6 is 12.

Watch for: Counting only one frame (6).; Writing 66.

S3. Fill in the missing numbers.

Answer: double 2 is 4; double 5 is 10; double 7 is 14; double 9 is 18

Double 2 is 2 + 2 = 4. Double 5 is 5 + 5 = 10. Double 7 is 7 + 7 = 14. Double 9 is 9 + 9 = 18.

Watch for: Writing the number again (double 7 is 7).; Adding 2 to the number (double 7 is 9).

S4. Look at the bars. Fill in the missing numbers.

Answer: 3 tens + 3 tens = 6 tens; 30 + 30 = 60; double 40 is 80

3 tens and 3 tens make 6 tens. 6 tens is 60. Double 4 tens is 8 tens, which is 80.

Watch for: Writing 6 for 30 + 30.; Writing 33 or 303.

S5. Ali has 8 stickers. Ben has 8 stickers. How many stickers do they have altogether?

Answer: 8 + 8 = 16; they have 16 stickers

Ali has 8 and Ben has 8. 8 + 8 = 16. They have 16 stickers altogether.

Watch for: Writing 8 (only one child).; Writing 88.

S6. Read each statement. Decide if it is True or False.

Answer: True, False, True, True

Double 4 is 8: True. Double 6 is 12, not 14: False. 9 + 9 is two of the same number, so it is a double: True. Double 30 is 60: True.

Watch for: Marking Double 6 is 14 as true because 6 + 6 is close to 14.; Saying 9 + 9 is not a double because 9 is a big number.

Think harder (page 3)

T1. Look at the counters. Fill in the missing numbers.

Answer: double 7 is 14; 7 + 8 = 15

The red frame is double 7 with the yellow: 7 + 7 = 14. The second frame has 8 yellow counters, one more than 7, so 7 + 8 = 14 + 1 = 15.

Watch for: Writing 14 for both parts.; Adding 2 instead of 1.

T2. Add.

Answer: 13, 17, 9, 19

6 + 7: double 6 is 12, and 1 more is 13. 8 + 9: double 8 is 16, and 1 more is 17. 4 + 5: double 4 is 8, and 1 more is 9. 9 + 10: double 9 is 18, and 1 more is 19.

Watch for: Taking 1 away instead of adding 1 after the smaller double.; Writing the double itself (12 for 6 + 7).

T3. Which sum is not a near double? Write the number in the bracket.

Answer: (3) 5 + 8

6 and 7 are next to each other. 8 and 9 are next to each other. 9 and 10 are next to each other. 5 and 8 are not next to each other (they are 3 apart), so 5 + 8 is not a near double.

Watch for: Choosing 9 + 10 because 10 looks different.; Looking at the total instead of the two numbers.

T4. Kumar adds 8 + 9. Step 1: 8 + 8 = 16. Step 2: 16 − 1 = 15. Check Step 2. Answer (a) and (b).

Answer: No; 8 + 9 = 17

9 is 1 more than 8, so the total is 1 more than 16: 16 + 1 = 17. Kumar took 1 away instead of adding 1.

Watch for: Saying Yes because 16 − 1 looks like a good step.; Writing 15 for the right answer.

Go further (page 4)

G1. Add 8 + 9 + 8 + 7 + 8. Show your working on the lines.

Answer: 40

The 9 is 1 more than 8 and the 7 is 1 less than 8, so they make up for each other. The five numbers are the same as five 8s: 8 + 8 + 8 + 8 + 8 = 40. Another way: 8 + 8 = 16, 9 + 7 = 16, and 16 + 16 + 8 = 40.

Watch for: Adding from left to right and slipping.; Leaving one number out.

G2. Ben says 9 + 10 + 11 is the same as 10 + 10 + 10. Answer (a) and (b). Show your working on the lines.

Answer: Yes; 19 + 20 + 21 = 60

9 is 1 less than 10 and 11 is 1 more than 10, so they make up for each other: 9 + 10 + 11 = 30 = 10 + 10 + 10. In the same way 19 + 20 + 21 is three 20s, which is 60.

Watch for: Saying No because the numbers look different.; Writing 50 for the three numbers.

G3. Find double 35. Fill in the missing numbers. Show your working on the lines.

Answer: 60, 10, 70

35 is 30 and 5. Double 30 is 60. Double 5 is 10. 60 + 10 = 70. So double 35 is 70.

Watch for: Writing 610 (joining 60 and 10).; Forgetting to double the 5.

Investigation (page 5): Make two groups the same

It works for 6 and 8 (both 7), and for 5 and 9 (both 7): the two numbers are 2 apart or 4 apart and the total is a double (14). It does not work for 7 and 8: the total is 15 and one dot is always left over.

Look for: A child who tries several pairs, moves dots, and notices that when the total is a double the two groups can be made the same, and that 7 and 8 always leave one dot over. Credit any clear reason. Do not use the words odd or even yet: that is the next lesson.

Exit ticket (page 5)

E1. Fill in the missing number.

Answer: 18

9 + 9 = 18. Move 1 from one 9 to the other: 10 and 8.

Watch for: Writing 9 or 11.; Writing 99.

E2. Fill in the missing number.

Answer: 60

Double 3 is 6, so double 3 tens is 6 tens: 60.

Watch for: Writing 6.; Writing 33 or 303.

E3. Add.

Answer: 17

Double 8 is 16 and 1 more is 17. Or double 9 is 18 and 1 less is 17.

Watch for: Writing 18 (the double of 9).; Writing 16.

Sneak-peek puzzles (not in the workbook)

1. 8
2. Yes. 6 + 7 is 6 + 6 with one more counter, so 12 + 1 = 13.